Fundamentals Chapter 9

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1a. $\overline{\overline{a+bi}} = \overline{a-bi} = a+bi$.

1b. $\overline{a+bi+c+di} = \overline{(a+c)+(b+d)i} = a-bi+c+di = \overline{a+bi} + \overline{c+di}$.

1c. $\overline{(a+bi)(c+di)} = \overline{(ac-bd)+(ad+bc)i} = (ac-bd)-(ad+bc)i = a(c-di)-bi(c-di) = (\overline{a+bi})(\overline{c+di})$.

1d. $|a-bi| = a^2+b^2 = |a+bi|$.

1e. $\overline{z^n} = \overline{z} \cdot \overline{z^{n-1}} = \cdots = \overline{z}^n$.

1f. If f is a polynomial, then

$$f(\overline{z}) = \sum_{i=0}^{n} a_i \overline{z}^i = \overline{\sum_{i=0}^{n} a_i z^i} = \overline{f(z)}.$$

We could also extend this to include power series, which would include functions like sin or cos that have a Taylor series.

2. $\{(1,0,0),(0,2,0),(0,0,3)\}$
$\{(4,0,0),(0,5,0),(0,0,6)\}$
$\{(7,0,0),(0,8,0),(0,0,9)\}$

3a. Let $\sum a_i \mathbf{w}_i = 0$ for some constants a_i. If we take the inner product on both sides with $\mathbf{w}_j$ for some $1 \leq j \leq k$, we have

$$\left\langle \mathbf{w}_j, \sum a_i \mathbf{w}_i \right\rangle = \langle \mathbf{w}_j, 0 \rangle = 0.$$

Since S is a set of orthonormal vectors, we have that $\langle \mathbf{w}_i, \mathbf{w}_j \rangle = 0$ for all $i \neq j$ and $\langle \mathbf{w}_i, \mathbf{w}_i \rangle = 1$. So

$$\left\langle \mathbf{w}_i, \sum a_i \mathbf{w}_i \right\rangle = a_j \langle \mathbf{w}_j, \mathbf{w}_j \rangle = a_j = 0.$$

Since j is arbitrary, all $a_i = 0$ where $1 \leq i \leq k$.

3b. Assume that S does not span V for contradiction. Then there exists $\mathbf{v} \in V$ that cannot be expressed as a linear combination of $\mathbf{w}_i$.

Consider the set $S' = S \cup \{\mathbf{v}\}$. Since $\mathbf{v}$ is not in the span of S, S' is still linearly independent. But V is an n-dimensional vector space, so the maximum number of linearly independent vectors in V is n.

Since S' has $n+1$ elements, this is a contradiction, so there exists no $\mathbf{v}$ that cannot be expressed as a linear combination of $\mathbf{w}_i$. Therefore, since S is a spanning set and is linearly independent, S forms a basis for V.

4a. By the definition of an inner product, $\langle \mathbf{u}, \mathbf{u}, \rangle > 0$ for nonzero $\mathbf{u}$, so $\|\mathbf{u}\| > 0$. If $\mathbf{u} = 0$, then we see that $\langle \mathbf{v}, 0\rangle + \langle -\mathbf{v}, 0\rangle = \langle \mathbf{v} - \mathbf{v}, 0\rangle = \langle 0, 0\rangle$, but also $\langle \mathbf{v}, 0\rangle + \langle -\mathbf{v}, 0\rangle = \langle \mathbf{v}, 0\rangle - \langle \mathbf{v}, 0\rangle = 0$, so $\langle 0, 0\rangle = 0$ and $\|\mathbf{0}\| = 0$.

4b. This follows from part c: $\left\| \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\| = \frac{1}{\|\mathbf{v}\|}\|\mathbf{v}\| = 1$.

4c. $\|\alpha\mathbf{v}\| = \sqrt{\langle \alpha\mathbf{v}, \alpha\mathbf{v}\rangle} = \sqrt{\alpha\overline{\alpha}\langle \mathbf{v}, \mathbf{v}\rangle} = |\alpha|\sqrt{\langle \mathbf{v}, \mathbf{v}\rangle} = |\alpha|\|\mathbf{v}\|$.

6. Since $A = (\mathbf{v}_1, \ldots, \mathbf{v}_m)$, we see that

$$\mathbf{v}_i = (a_{i1}, \ldots, a_{im}).$$

Similarly,

$$\mathbf{w}_j = (b_{1j}, \ldots, b_{mj}).$$

Taking the inner product of $\mathbf{v}_i$ and $\overline{\mathbf{w}_j}$, we get

$$\langle \mathbf{v}_i, \mathbf{w}_j \rangle = \sum_{\ell=1}^{m} a_{i\ell}\overline{\overline{b_{\ell j}}} = \sum_{\ell=1}^{m} a_{i\ell} b_{\ell j},$$

which is the same expression that defines c_{ij}. So $c_{ij} = \langle \mathbf{v}_i, \overline{\mathbf{w}_j}\rangle$.

7.

$$\mathbf{v}_1 = (1,1,0), \quad \mathbf{v}_2 = (0,1,1)$$

$$\mathbf{u}_1 = (1,1,0), \quad \mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1}\mathbf{v}_2$$

$$\text{proj}_{\mathbf{u}_1}\mathbf{v}_2 = \frac{\langle \mathbf{v}_2, \mathbf{u}_1\rangle}{\langle \mathbf{u}_1, \mathbf{u}_1\rangle}\mathbf{u}_1 = \frac{1}{2}(1,1,0) = \left(\frac{1}{2}, \frac{1}{2}, 0\right)$$

$$\mathbf{u}_2 = (0,1,1) - \left(\frac{1}{2}, \frac{1}{2}, 0\right) = \left(-\frac{1}{2}, \frac{1}{2}, 1\right)$$

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1,0) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0\right)$$

$$\mathbf{e}_2 = \frac{1}{\sqrt{3/2}}\left(-\frac{1}{2}, \frac{1}{2}, 1\right) = \left(-\frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{6}, \frac{\sqrt{6}}{3}\right).$$

8. We have that $\mathbf{e}_1 = (1,0), \mathbf{e}_2 = (0,1), \mathbf{e}_1 + \mathbf{e}_2 = (1,1)$. Let $\mathbf{v} = (v_1, v_2)$. We see that

$$\frac{\sum |\langle \mathbf{v}, \mathbf{w} \rangle|^2}{\langle \mathbf{v}, \mathbf{v} \rangle} = \frac{v_1^2 + v_2^2 + (v_1 + v_2)^2}{v_1^2 + v_2^2} = \frac{2v_1^2 + 2v_2^2 + 2v_1 v_2}{v_1^2 + v_2^2} = 2 + \frac{2v_1 v_2}{v_1^2 + v_2^2}.$$

Our goal is to find $0 < A \leq B$ such that

$$A \leq 2 + \frac{2v_1 v_2}{v_1^2 + v_2^2} \leq B$$

for all $v_1, v_2 \in \mathbb{R}$.

We know that $(v_1 + v_2)^2 \geq 0$. Rearranging this gives us $\frac{2v_1 v_2}{v_1^2 + v_2^2} \geq -1$, so $2 + \frac{2v_1 v_2}{v_1^2 + v_2^2} \geq 1$. This means we can let $A = 1$.

Also, we know that $(v_1 - v_2)^2 \geq 0$. Rearranging gives us $\frac{2v_1 v_2}{v_1^2 + v_2^2} \leq 1$, so we can let $B = 3$.

9a. We want to show that for all $\mathbf{v} \in \mathbb{C}^2$,

$$A\langle \mathbf{v}, \mathbf{v} \rangle \leq \sum |\langle \mathbf{v}, \mathbf{w} \rangle|^2 \leq B\langle \mathbf{v}, \mathbf{v} \rangle.$$

Let $\mathbf{v} = (c_1, c_2)$ We see that

$$\sum |\langle \mathbf{v}, \mathbf{w} \rangle|^2 = 2|\langle \mathbf{v}, \mathbf{e}_1 \rangle|^2 + |\langle \mathbf{v}, \mathbf{e}_2 \rangle|^2 = 2|c_1|^2 + |c_2|^2.$$

Also, we see that $\langle \mathbf{v}, \mathbf{v} \rangle = c_1\overline{c_1} + c_2\overline{c_2} = |c_1|^2 + |c_2|^2$. So our goal is to find $0 < A \leq B$ such that

$$A \leq \frac{2|c_1|^2 + |c_2|^2}{|c_1|^2 + |c_2|^2} \leq B.$$

Since $\frac{2|c_1|^2 + |c_2|^2}{|c_1|^2 + |c_2|^2} = 1 + \frac{|c_1|^2}{|c_1|^2 + |c_2|^2}$ and $0 \leq \frac{|c_1|^2}{|c_1|^2 + |c_2|^2} \leq 1$, we can let $A = 1$ and $B = 2$.

9b. Let $\mathbf{v} = (c_1, c_2)$ where $c_1 = a + bi$ and $c_2 = c + di$. Then we have

$$\frac{1}{\langle \mathbf{v}, \mathbf{v} \rangle} \sum |\langle \mathbf{v}, \mathbf{w} \rangle|^2 = \frac{|c_1|^2 + |c_2|^2 + |c_1 + c_2|^2 + |c_1 - c_2|^2}{|c_1|^2 + |c_2|^2}$$

$$= 1 + \frac{|c_1 + c_2|^2 + |c_1 - c_2|^2}{|c_1|^2 + |c_2|^2} = 1 + \frac{(a+c)^2 + (b+d)^2 + (a-c)^2 + (b-d)^2}{a^2 + b^2 + c^2 + d^2}$$

$$= 1 + \frac{2a^2 + 2b^2 + 2c^2 + 2d^2}{a^2 + b^2 + c^2 + d^2} = 3.$$

So we can let $A = B = 3$.